

Deepak Sathyan

Curriculum Vitae

[arXiv](#) | [InspireHEP](#) | [Google Scholar](#) | [ORCID](#) | [Website](#)

✉ dsathyan@tamu.edu

📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Experience

Postdoctoral Research Associate

Mitchell Institute for Fundamental Physics and Astronomy
Texas A&M University

College Station, TX
Sept 2024 – Aug 2027
3 years

Education

University of Maryland

Ph.D. in Physics

Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass
Advisor: Dr. Kaustubh Agashe

College Park, MD
Aug 2018 – July 2024

Boston University

Bachelor of Arts in Physics, Summa Cum Laude with Honors

GPA: 3.9
Minor: Mathematics
Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment
Advisor: Dr. Robert Carey

Boston, MA
Aug 2014 – May 2018

Research Interests

Beyond the Standard Model searches with Collider Physics: LHC and muon collider phenomenology, Randall-Sundrum and baryon-number-violation models

Searches for sub-GeV Dark Matter: dark matter-neutrino interactions, dark photon and dark sector production, cosmic ray-dark matter scattering

Neutrino Physics: supernova neutrino attenuation and detection, open-system observables, neutrinophilic mediators, neutrino NSI

Publications

5. New laboratory constraints on neutrinophilic mediators

P.S.B. Dev, D. Kim, **D. Sathyan**, K. Sinha, Y. Zhang
arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)

July 2024

4. ‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W

K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, **D. Sathyan**
arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)

Apr 2024

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| <p>3. A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET
 K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
 arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)</p> | <p>Oct 2023</p> |
| <p>2. Energy-peak based method to measure top quark mass via B-hadron decay lengths
 K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
 arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)</p> | <p>Dec 2022</p> |
| <p>1. LHC Signals for KK Graviton from an Extended Warped Extra Dimension
 K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan
 arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)</p> | <p>Aug 2020</p> |

Preprints

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| <p>4. Intensity-Frontier Signals of Warped Extra Dimensions
 D. Kim, D. Sathyan, A. Verma
 arxiv.org/abs/2608.20482</p> | <p>Aug 2026</p> |
| <p>3. Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering
 B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan
 arxiv.org/abs/2605.08010</p> | <p>May 2026</p> |
| <p>2. A Baryon and Lepton Number Violation Model Testable at the LHC
 A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
 arxiv.org/abs/2508.21064</p> | <p>Aug 2025</p> |
| <p>1. New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis
 P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
 arxiv.org/abs/2507.01000</p> | <p>July 2025</p> |

Articles in Preparation

- 1. LHC as a Beam Dump: Probing Light Dark Photons**
B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, **D. Sathyan**
(In preparation)
- 2. New Probes of Dark Sectors at a Muon Collider**
B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, **D. Sathyan**
(In preparation)
- 3. Exploring open systems approach to neutrino oscillations**
A. Chandra Shekar, K. Kelly, **D. Sathyan**, L. Strigari, T. Zhou
(In preparation)

4. LHC Signals for KK Gravitons Producing 8-Particle Final States

K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, **D. Sathyan**
(In preparation)

Snowmass2021 Contributions

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| 5. TF07 Snowmass Report: Theory of Collider Phenomena
F. Maltoni et al.
arxiv.org/abs/2210.02591 | Oct 2022 |
| 4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021
K. Agashe et al.
arxiv.org/abs/2209.11267 | Sept 2022 |
| 3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021
T. Bose et al.
arxiv.org/abs/2209.13128 | Sept 2022 |
| 2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond
K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan
arxiv.org/abs/2204.02928 | Apr 2022 |
| 1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models
K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan
arxiv.org/abs/2203.13305 | Mar 2022 |

Muon g-2 Articles

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| 3. The straw tracking detector for the Fermilab Muon g-2 Experiment
B.T. King et al.
arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035) | Nov 2021 |
| 2. Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab
Muon g-2 Collaboration
arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002) | Apr 2021 |
| 1. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm
Muon g-2 Collaboration
arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801) | Apr 2021 |

Ph.D. Thesis

- 1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)

Seminars

- 9. HEP Seminar at Northwestern University** Sept 2026
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess
- 8. Fermilab Cosmic Physics Center Seminar** Sept 2026
Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess
- 7. CMS Exotica general meeting** May 2024
Unification of Searches and Measurements: Probing BSM with the W boson mass measurement
- 6. Seminar at Texas A&M** Dec 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 5. Seminar at UT Austin** Dec 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 4. CMS SUSY general meeting** Nov 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 3. HEP Seminar at University of Pittsburgh** Nov 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 2. HEP Seminar at Northwestern University** Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
- 1. Theory Seminar at Washington University in St. Louis** Oct 2023
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

Conference Talks

- 14. Light Dark World 2026 at Carleton University [Invited]** July 2026
Can the LHC be sensitive to the Light Dark World?
- 13. CETUP* Workshop [Invited]** June 2026
Producing the GeV Galactic Center Excess via Cosmic Ray–Dark Matter Scattering

12. Particle Physics on the Plains at the University of Kansas [Contrib.] A Baryon and Lepton Number Violation Model Testable at the LHC	Nov 2025
11. CETUP* Workshop [Invited] Can the LHC be sensitive to light dark mediators?	June 2025
10. Phenomenology Conference at the University of Pittsburgh [Contrib.] Can the LHC be sensitive to light dark mediators?	May 2025
9. Particle Physics on the Plains at the University of Kansas [Contrib.] A comprehensive analysis of supernova neutrino-dark matter interactions	Nov 2024
8. Texas TACOS @ UT Austin [Invited] A comprehensive analysis of supernova neutrino-dark matter interactions	Oct 2024
7. NuFact 2024 Conference at Argonne National Laboratory [Invited] A comprehensive analysis of supernova neutrino-dark matter interactions	Sept 2024
6. Mitchell Conference at Texas A&M University [Invited] A comprehensive analysis of supernova neutrino-dark matter interactions	May 2024
5. Particle Physics on the Plains at the University of Kansas [Contrib.] A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET	Oct 2023
4. Phenomenology Conference at the University of Pittsburgh [Contrib.] Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos	May 2023
3. Phenomenology Conference at the University of Pittsburgh [Contrib.] Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)	May 2022
2. April APS Meeting [Contrib.] Signals of KK graviton from extended warped extra dimensions at the LHC (II)	Apr 2021
1. Phenomenology Conference at the University of Pittsburgh [Contrib.] Signals of KK graviton from extended warped extra dimensions at the LHC (II)	May 2020

Teaching

Principles of Modern Physics Teaching Assistant, University of Maryland	Spring 2024
Electricity and Magnetism I Teaching Assistant, University of Maryland	Spring 2024
Mathematical Methods for Physics II Teaching Assistant, University of Maryland	Fall 2021, Fall 2022
Fundamentals of Physics I Teaching Assistant, University of Maryland	Summer 2019, Spring 2022

Introduction to Quantum Mechanics I (Graduate) Teaching Assistant, University of Maryland	Fall 2020
Fundamentals of Physics for Biologists II Teaching Assistant, University of Maryland	Spring 2019, Spring 2020
Fundamentals of Physics for Biologists Teaching Assistant, University of Maryland	Fall 2018, Fall 2019
Guest Lecture for Undergraduate E&M at University of Maryland Lectured on vector potentials in E&M	Mar 2024
Guest Lecture for Graduate E&M at Texas A&M University Lectured on relativistic Larmor formula and charged particle motion in magnetic fields	Oct 2025

Service

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026 Conference Organizer — Texas A&M University, May 27-30, 2026
The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025 Conference Organizer — Texas A&M University, May 13-16, 2025
Mitchell Institute High Energy Phenomenology, Experiment, and Cosmology Seminar Series Weekly Seminar Organizer - Texas A&M University, Fall 2024 - Present

Awards

Breakthrough Prize in Fundamental Physics Awarded to the Muon g-2 Collaboration; recognized as a collaboration member	Apr 2026
Ralph Myers & Friends of Physics Award Awarded by the University of Maryland Department of Physics	Dec 2019
Ralph Myers & Friends of Physics Award Awarded by the University of Maryland Department of Physics	May 2020
Alumni Award Awarded by the Boston University Department of Physics	May 2018

Schools Attended

GGI Lectures on the Theory of Fundamental Interactions	Jan 2023
TASI 2024: The Frontiers of Particle Theory	June 2024

Outreach

BU Student Mentorship Program PRISM

2016 - 2018

Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

BU Outreach for Local High Schools

Jan 2018

Demonstrated measurement of the charge-to-mass ratio of the electron for local high schoolers in physics

TAMU Physics and Engineering Festival

2025 - present

Presented demonstrations of fluid dynamics to the general public, including high school students and families

University of South Dakota talk: "My path through particle physics research so far"

Dec 2024

Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

Skills

Programming:

Mathematica, Python, C++

Physics Software:

MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules

References

Dr. Kaustubh Agashe

University of Maryland, College Park — kagashe@umd.edu

Dr. Bhaskar Dutta

Texas A&M University — dutta@tamu.edu

Dr. Bhupal Dev

Washington University in St. Louis — bdev@wustl.edu

Dr. Jason Kumar

University of Hawai'i — jkumar@hawaii.edu